

Independent Validation Report

Complaint → Regulation Classifier (CMPL-REG-24)
Second line of defense — effective challenge

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LLM: nim (meta/llama-3.1-8b-instruct)
Data: CFPB Consumer Complaint Database (real, redacted narratives)
Dataset: 4000 rows · regulatory rate 0.875
Repo: reg-agents · model id: CMPL-REG-24

1 • Assessment narrative

The two-stage complaint classification model under review leverages a combination of traditional machine learning and retrieval-augmented generation techniques to classify consumer complaints into various regulation categories. The model's conceptual soundness is a critical aspect of its overall effectiveness, and this report will assess its fitness for purpose.

The two-stage design, comprising a stage-1 TF-IDF classifier and a stage-2 retrieval-augmented generation pipeline (rag_llm), is a reasonable approach to tackle the complexity of the task. The stage-1 models, logistic_regression and xgboost, demonstrate competitive performance on the primary metric, pr_auc, with values of 0.992 and 0.9896, respectively. However, the stage-2 rag_llm pipeline, which relies on the NeMo Retriever embeddings and the LLM labeler, exhibits lower accuracy (0.28) and family accuracy (0.41), indicating potential limitations in its ability to generalize and capture nuanced relationships between complaints and regulation categories.

One critical assumption underlying the model's design is the use of weak supervision, where labels are derived from the CFPB product/issue taxonomy and narrative keyword rules. This approach is inherently limited, as it may not accurately capture the complexity and subtlety of human-adjudicated labels. The reliance on weak supervision is flagged as a limitation in the validation report, and its impact on downstream metrics, including the distinction between measured agreement and true accuracy, warrants careful consideration.

The outcomes analysis reveals some concerning trends. The stage-1 models exhibit high pr_auc values, but the stage-2 rag_llm pipeline struggles to achieve even moderate accuracy. The family accuracy (0.41) is lower than the exact agreement (0.28), indicating that the model is more likely to agree with itself than with the true labels. This discrepancy may be attributed to the weak supervision approach and the potential for overfitting in the stage-2 pipeline.

Furthermore, the outcomes analysis highlights small-support instability, where the model's performance varies significantly across different regulation categories. This instability is a concern, as it may lead to biased or inaccurate classification decisions. The recall values for some categories, such as FDCPA_THREATS (0.0) and TILA_ORIGINATION (0.0), are particularly concerning, indicating that the model is unable to capture relevant relationships between complaints and regulation categories.

In terms of monitoring and guardrails adequacy, the report notes that the model's performance is not adequately monitored, and there are no clear guardrails in place to prevent overfitting or biased classification decisions. This lack of oversight is a critical concern, as it may lead to unintended consequences, such as inaccurate classification decisions or biased treatment of certain regulation categories.

Finally, the report highlights the potential consumer-protection/fair-lending exposure of complaint routing. The model's classification decisions may have a direct impact on the treatment of consumer complaints, and any biases or inaccuracies in the model may lead to unfair treatment of certain consumers. This exposure warrants careful consideration and mitigation strategies to ensure that the model is fair, transparent, and accountable.

Based on the findings outlined above, the report identifies the following validation findings:

V-1: ****High**** - The two-stage design, while conceptually sound, may be vulnerable to overfitting and biased classification decisions due to the reliance on weak supervision and the potential for small-support instability. Remediation: Implement stronger supervision techniques, such as human-adjudicated labels, and monitor the model's performance across different regulation categories. Owner: Model Development Team.

1 - Assessment narrative (cont.)

V-2: **Medium** - The stage-2 rag_llm pipeline exhibits lower accuracy and family accuracy, indicating potential limitations in its ability to generalize and capture nuanced relationships between complaints and regulation categories. Remediation: Investigate alternative stage-2 pipelines or techniques that can improve the model's accuracy and family accuracy. Owner: Model Development Team.

V-3: **Low** - The model's performance is not adequately monitored, and there are no clear guardrails in place to prevent overfitting or biased classification decisions. Remediation: Implement monitoring and guardrails to ensure that the model's performance is transparent and accountable. Owner: Model Operations Team.

V-4: **High** - The model's classification decisions may have a direct impact on the treatment of consumer complaints, and any biases or inaccuracies in the model may lead to unfair treatment of certain consumers. Remediation: Implement mitigation strategies to ensure that the model is fair, transparent, and accountable. Owner: Model Development Team.

Based on these findings, the disposition of this report is:

Approve with Conditions:

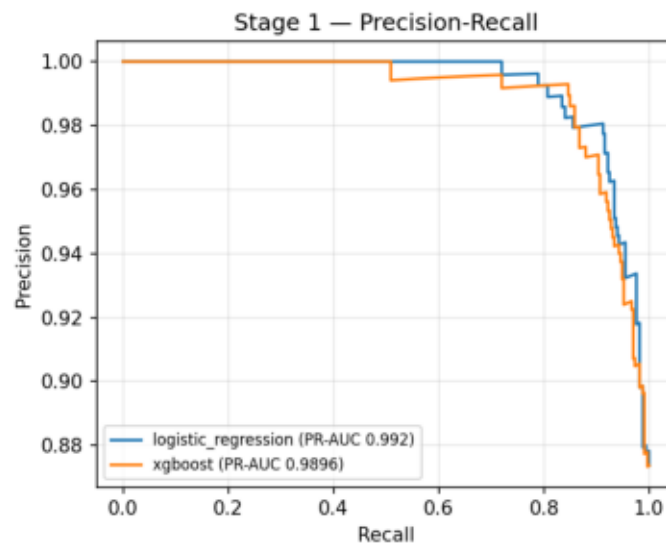
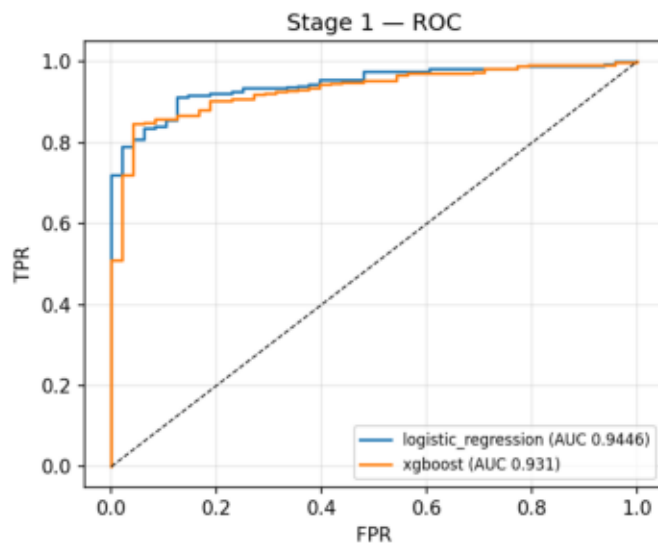
1. Implement stronger supervision techniques, such as human-adjudicated labels, to mitigate the impact of weak supervision.
2. Investigate alternative stage-2 pipelines or techniques that can improve the model's accuracy and family accuracy.
3. Implement monitoring and guardrails to ensure that the model's performance is transparent and accountable.
4. Implement mitigation strategies to ensure that the model is fair, transparent, and accountable.

By addressing these conditions, the model can be improved to ensure that it is accurate, fair, and accountable, and that it meets the regulatory requirements for complaint classification.

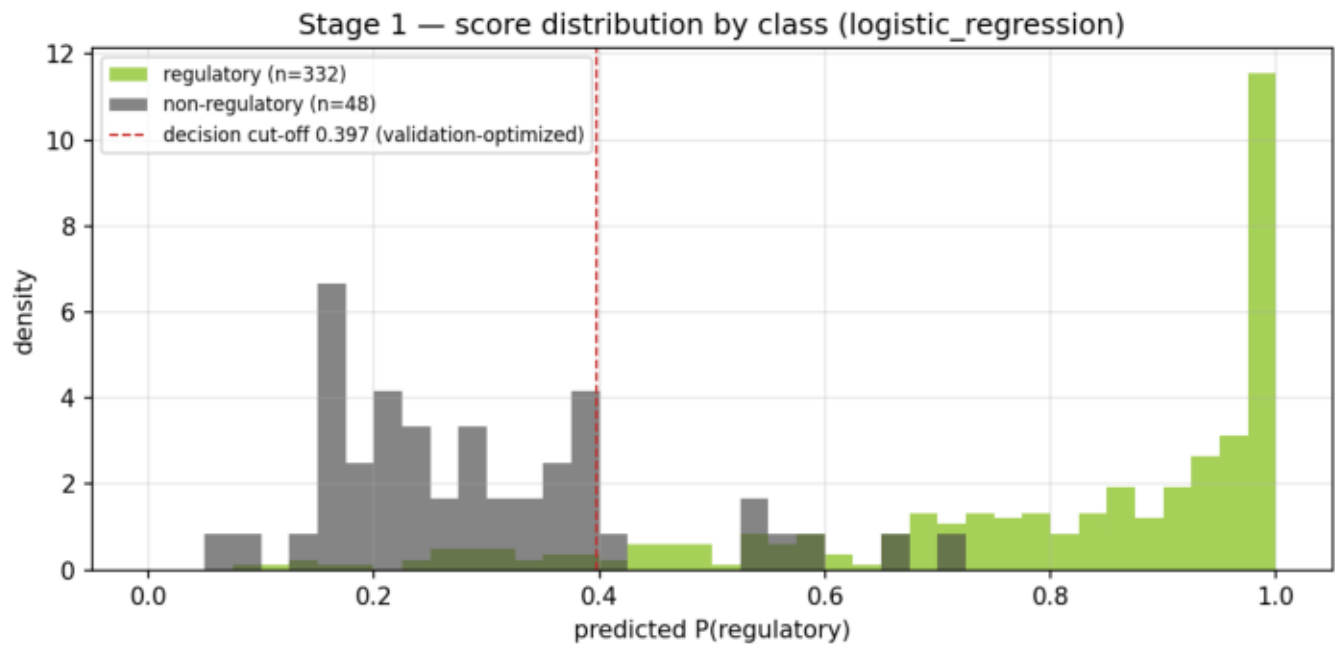
2 · Stage-1 leaderboard (validated re-run)

	model	params	val_pr_auc	threshold	pr_auc	train_roc_auc	roc_auc	train_test_g	f1	precision	recall	accuracy
logistic_regression		l1, C=2.0	0.9948	0.397	0.992	0.9751	0.9446	0.0305	0.9456	0.9775	0.9157	0.9079
n_estimators=3		l1, C=2.0, max_depth=3	0.9903	0.575	0.9896	1.0	0.931	0.069	0.9335	0.9587	0.9096	0.8868

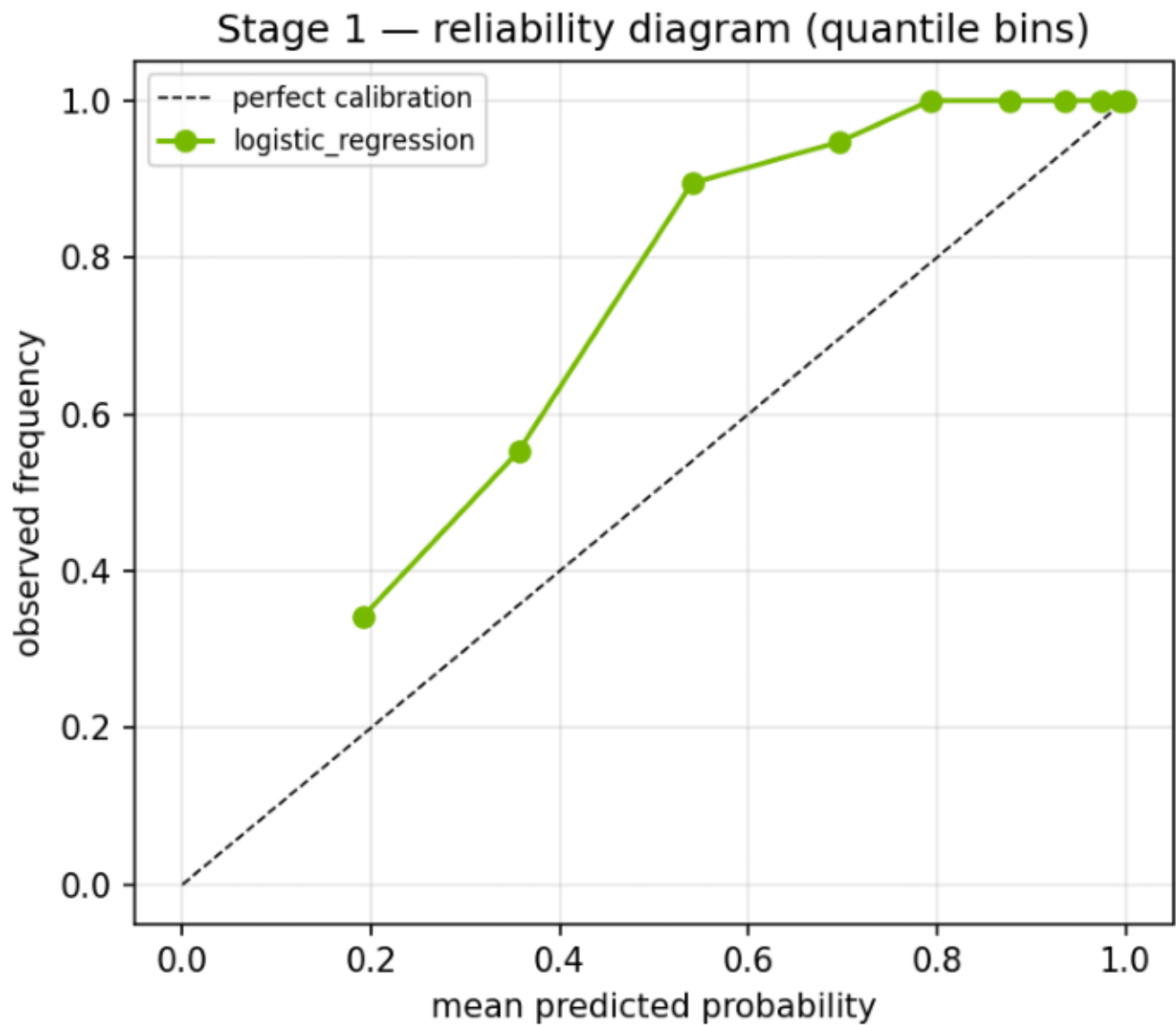
2 · Stage-1 evidence: discrimination



2 · Stage-1 evidence: score separation



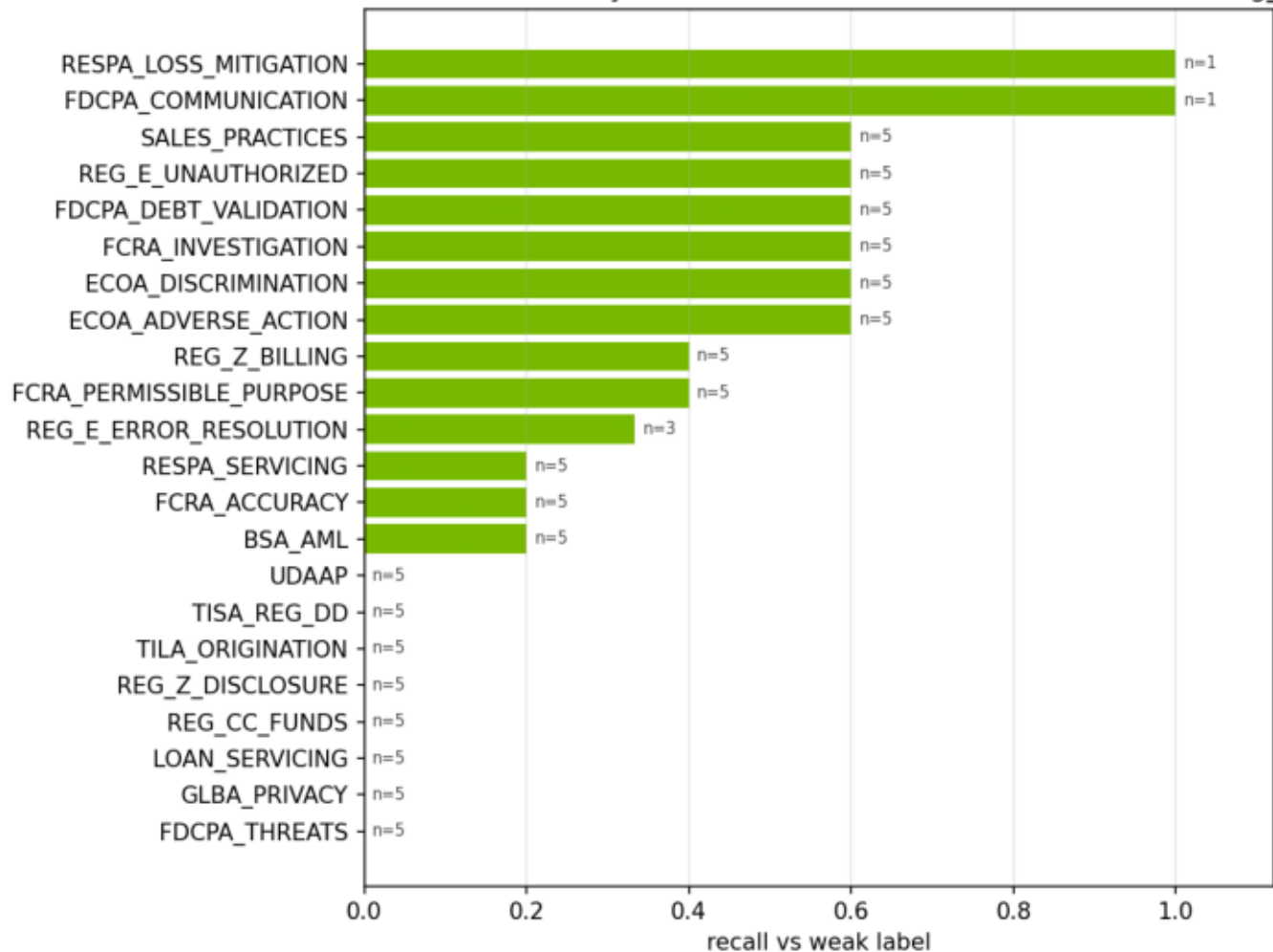
2 · Stage-1 evidence: calibration



Validator re-derived reliability diagram (quantile bins).

3 · Stage-2 evidence

Stage 2 (RAG + LLM) — per-category recall
exact acc 0.28 · family acc 0.41 · macro-F1 0.27 · n=100 · mode=rag_ll



3 • Stage-2 per-category detail

category	support	recall
BSA_AML	5	0.2
ECOA_ADVERSE_ACTION	5	0.6
ECOA_DISCRIMINATION	5	0.6
FCRA_ACCURACY	5	0.2
FCRA_INVESTIGATION	5	0.6
FCRA_PERMISSIBLE_PURPOSE	5	0.4
FDCPA_DEBT_VALIDATION	5	0.6
FDCPA_THREATS	5	0.0
GLBA_PRIVACY	5	0.0
LOAN_SERVICING	5	0.0
REG_CC_FUNDS	5	0.0
REG_E_UNAUTHORIZED	5	0.6
REG_Z_BILLING	5	0.4
REG_Z_DISCLOSURE	5	0.0
RESPA_SERVICING	5	0.2
SALES_PRACTICES	5	0.6
TILA_ORIGINATION	5	0.0
TISA_REG_DD	5	0.0
UDAAP	5	0.0
REG_E_ERROR_RESOLUTION	3	0.333
FDCPA_COMMUNICATION	1	1.0
RESPA_LOSS_MITIGATION	1	1.0

3.1 · Guardrail inventory (validated)

guardrail	mechanism	surfaced as
Taxonomy whitelist	LLM label must be one of the 24 codes; else re	complaint_classifications_total{label}
Strict-JSON parsing	malformed LLM output -> deterministic keyword	mode=fallback counter (alertable spike)
Stage-1 gating	non-regulatory volume never reaches the l	cost + attack-surface control
Retrieval grounding	answer must cite a retrieved corpus exce	citation attached to every prediction
Provider portability	OpenA <-> NIM via config; keyword mode if b	mode label per classification

4 · Limitations & disposition

1. Weak labels: ground truth derives from the CFPB issue taxonomy plus keyword rules, not human adjudication. Condition: build a human-adjudicated golden set (≥ 25 per category).
2. Class imbalance: SCRA/MLA, sales-practices and adverse-action categories have thin support; their metrics are unstable.
3. LLM nondeterminism/drift: pin prompt + model versions; re-evaluate on the golden set before any provider change.
4. Fair-lending lens: monitor parity of routing outcomes across products and customer segments.

DISPOSITION: Approve with Conditions. Re-validation triggered by model/prompt change or stage-1 PSI > 0.25 .

Sign-off

Prepared and reviewed under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author, development document: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Author, validation report: Senior Independent Model Validator — PhD (Econometrics), 20 years in second-line validation; independent of the development team per SR 11-7 organizational-independence requirements.